BROWN REPRESENTABILITY IN A GROTHENDIECK ABELIAN CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the single omitted step is the sentence 'This construction gives an inverse to the map ... (details omitted).', a routine verification that the map just constructed is inverse to a displayed map. It is preserved verbatim, and the whole construction leading to it is written out in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The functor represented by an object. Same chapter, the discussion of the functor represented by an object and its universal element. Supplies the description of a representable functor and the universal element used in the construction.
- The small object argument in the form used here. Same chapter, Lemma proposition-modules-are-small and the surrounding discussion of smallness with respect to injections. Supplies that the objects of a Grothendieck abelian category are small with respect to injections, which produces the subobject generated by the image of a given element.
- Colimits and their universal property. Same chapter, the universal property of colimits in the category. Supplies the direction in which representable functors commute with colimits and the description of colimits used to build the representing object.

Source reference. Original section 010M: sources/stacks-homology/pinned-original-source.tex, source lines 1680–1680. Navigation pointer invoked at proof line 2140; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let $F : \mathcal{A}^{opp} \to Sets$ be a functor. Then F is representable if and only if F commutes with colimits, i.e.,*

$$F(\operatorname{colim}_i N_i) = \lim F(N_i)$$

for any diagram $\mathcal{I} \to \mathcal{A}$, $i \in \mathcal{I}$.

Proof. If F is representable, then it commutes with colimits by definition of colimits.

Assume that F commutes with colimits. Then $F(M \oplus N) = F(M) \times F(N)$ and we can use this to define a group structure on $F(M)$. Hence we get $F : \mathcal{A} \to Ab$ which is additive and right exact, i.e., transforms a short exact sequence $0 \to K \to L \to M \to 0$ into an exact sequence $F(K) \leftarrow F(L) \leftarrow F(M) \leftarrow 0$ (compare with Homology, Section 010M).

Let U be a generator for $\mathcal{A}$. Set $A = \bigoplus_{s \in F(U)} U$. Let $s_{univ} = (s)_{s \in F(U)} \in F(A) = \prod_{s \in F(U)} F(U)$. Let $A' \subset A$ be the largest subobject such that s_{univ} restricts to zero on A'. This exists because $\mathcal{A}$ is a Grothendieck category and because F commutes with colimits. Because F commutes with colimits there exists a unique element $\overline{s}_{univ} \in F(A/A')$ which maps to s_{univ} in $F(A)$. We claim that A/A' represents F, in other words, the Yoneda map

$$\overline{s}_{univ} : h_{A/A'} \longrightarrow F$$

is an isomorphism. Let $M \in \mathrm{Ob}(\mathcal{A})$ and $s \in F(M)$. Consider the surjection

$$c_M : A_M = \bigoplus\nolimits_{\varphi \in \mathrm{Hom}_{\mathcal{A}}(U,M)} U \longrightarrow M.$$

This gives $F(c_M)(s) = (s_\varphi) \in \prod_\varphi F(U)$. Consider the map

$$\psi : A_M = \bigoplus\nolimits_{\varphi \in \mathrm{Hom}_{\mathcal{A}}(U,M)} U \longrightarrow \bigoplus\nolimits_{s \in F(U)} U = A$$

which maps the summand corresponding to φ to the summand corresponding to s_φ by the identity map on U. Then s_{univ} maps to $(s_\varphi)_\varphi$ by construction. in other words the right square in the diagram

$$\begin{array}{ccccc} A' & \longrightarrow & A & \overset{s_{univ}}{\dashrightarrow} & F \\ {\scriptstyle ?}\uparrow & & \uparrow{\scriptstyle \psi} & & \uparrow{\scriptstyle s} \\ K & \longrightarrow & A_M & \longrightarrow & M \end{array}$$

commutes. Let $K = \mathrm{Ker}(A_M \to M)$. Since s restricts to zero on K we see that $\psi(K) \subset A'$ by definition of A'. Hence there is an induced morphism $M \to A/A'$. This construction gives an inverse to the map $h_{A/A'}(M) \to F(M)$ (details omitted). $\square$